

Investigating The Accuracy of Monte Carlo Simulation & Mie Theory, in The Context of Light Propagation in Whole Milk

April, 1,2025

Research Final Project, Intermediate Computational Methods

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Abstract

This study examined the reliability of light scattering theory in a turbid medium, specifically whole milk. The model simulated light propagation in a cube using Monte Carlo ray tracing and Mie theory. The flux through each face of the cube was calculated and then compared with measured intensities from a whole milk scattering sample. The globules were assumed to be isotropic spheres, which allowed the use of the Henyey Greenstein phase function. The results suggest that Monte Carlo ray tracing, together with the isotropic sphere approximation for milk, gives acceptable agreement with experimental observations, with a fitting uncertainty of approximately 8%. The dominant experimental uncertainty arose from Fresnel reflection, along with additional inelastic scattering effects. This model was evaluated using a 650 nm laser.

Introduction

Despite extensive research on photon transport, including studies of light interaction in dense media, sunscreen sprays, and blood vessels,[3] [2][4] accurately modeling light propagation in complex materials remains challenging. This difficulty arises from anisotropic scattering, polarization effects, and possible interference. This study addresses part of that gap by using a Monte Carlo simulation to model the random walk of photons through several media, beginning with milk and eventually extending to biologically relevant tissues such as cataract affected lenses in the eye. The experiment involved passing light through a cuvette and comparing the electromagnetic intensity measured on each face of the cuvette with the model predictions. The goal is to quantify the effective scattering properties of these media in both simulation and experiment. This work aims not only to improve understanding of photon material interactions, but also to provide groundwork for future applications in ophthalmology. The following sections present the methodology, results, and implications of the study, with an emphasis on light scattering in biological tissues and related systems.

Assumptions and Limitations

This model focuses on conditions in which Mie scattering is dominant and absorption is minimal, in order to simplify the experimental procedure. For room temperature milk over wavelengths of roughly 500 nm to 700 nm, the model does not include Rayleigh scattering from casein protein polymers, since that contribution is not dominant in this range.

Experimental limitations include uncertainty in the measured signal due to Fresnel reflection within the cuvette material before or after interaction, which accounts for roughly 4% of the traveling photons.

This experiment considers diluted whole milk with an x parts per drops milk to water ratio in order to demonstrate forward scattering effectively while maintaining a manageable number of interactions within the material for computational efficiency.

The computational model relies on a combination of Mie theory calculations from the classic text[1, 278 281] *Absorption and Scattering of Light by Small Particles*. The original function was translated to Fortran by Bruce T. Draine and later into Python by Clément Baruteau[8]. From there, the Henyey Greenstein function is applied as an approximation of the interaction event, using g , which can be interpreted as a measure of scattering anisotropy.

We assume that the fat globules are homogeneous spheres, which simplifies the interaction for the purposes of Mie theory calculations.

The experiment uses a sensitive photodetector to measure the intensity of light, but not all photons emitted from the cuvette surface are necessarily detected. This limitation is addressed by analyzing ratio distributions, which provides a reasonable comparison even though it is not a perfect measurement method.

Theory of light propagation in the context of the project

To begin, the assumption that light travels at a fixed speed inside a material must be set aside in favor of a microscopic view of how light interacts with the components that make up a material at the macroscopic level. To understand why such assumptions are valid in an effective sense, the mathematics describing these macroscopic interactions, specifically in milk, must begin with Mie theory.

Mie Theory

Mie theory, or Mie scattering, is most directly described through Maxwell's equations and the interaction of an electromagnetic plane wave with a homogeneous sphere. In this investigation, scattering and extinction cross sections were used, and these cross sections were represented as coefficients for the relevant interactions. The coefficients Q_a , Q_s , and Q_e correspond to the absorption coefficient, scattering coefficient, and extinction coefficient, respectively. These values are obtained through a series of summations involving spherical harmonics, which allow the coefficients to be computed.

$$Q_s = \frac{2}{k^2 r^2} \sum_{n=1}^{\infty} (2n+1) (|a_n|^2 + |b_n|^2) \quad (1)$$

Here, k is the spatial frequency and r is the radius of the particle. The terms a_n and b_n represent contributions from the electric and magnetic multipole interactions.

$$Q_e = \frac{2}{k^2 a^2} \sum_{n=1}^{\infty} (2n+1) \operatorname{Re}(a_n + b_n) \quad (2)$$

From these series, the scattering and extinction coefficients for a single interaction can be determined and then applied over a distance of 1 millimeter in the step size calculation.

Afflicted Milk Composition

For this experiment, the focus is on fat globule scattering in milk, which corresponds to Mie scattering rather than Rayleigh scattering from casein polymers. The composition of milk fat globules is relatively homogeneous and spherical. Therefore, approximating the fat globules as spheres is reasonable for this model, although some error is expected.

Objective

The objective of this project was to demonstrate, through computational modeling, how light scattering in turbid media such as milk can be approximated using Monte Carlo simulation and Mie theory calculations. The model was then compared with experimental results to determine whether it provided a good fit to real world behavior.

Experimental Set up

For the experimental setup, a 650 nm laser with output greater than 1 mW was aligned with a plastic cuvette. The sample cuvette was filled with water, and milk was added drop by drop and mixed thoroughly before measuring the intensity at each face of the cuvette with a photodiode, model s1723 05, connected to a power supply and voltmeter.

More information can be found here

<https://github.com/Hargis-Studies/Milk-Lab-Detailed-Procedure>

Methodology

To begin, I will describe the methodology of this project layer by layer. I will start with the more formulaic components and then move toward the underlying theory. The first point to note is that, because specialized equipment for measuring the absorption index of the specific whole milk sample was not available, values were taken from previous studies by [7] and [6] to estimate the absorption coefficient for raw milk.

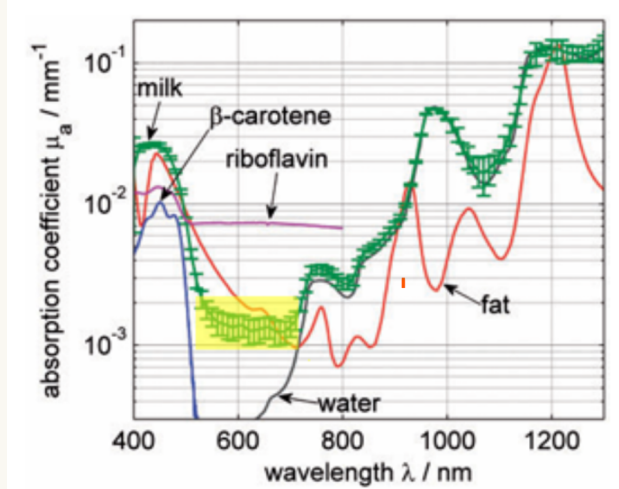


Figure 1: Absorption Coefficient by Wavelength, Figure 8.
Imaginary part of the refractive index (absorbance index)
for milk from 400 1300 nm. Data digitized and adapted from Stocker et al. (2016).

Moving on from the absorption coefficient, there are several quantities needed to predict light scattering in milk. These include the step size, the allowed values of θ and ϕ for scattering, the milk concentration, and the size of the domain under consideration. These measurements can be determined using the methods below.

Monte Carlo

To describe Monte Carlo in its simplest form, it can be thought of as randomness. Given a range of x possible points in space, one point is chosen, and the process is repeated.

In this project, the Beer Lambert law is used:

$$P = e^{-s\mu_t}$$

where μ_t is the extinction coefficient and s is the step size taken by the light before it encounters another milk particle. In a classic Monte Carlo simulation, since P lies between 0 and 1, we can treat it as a random variable, let p be some $\epsilon \in [0, 1]$, and solve for the sampled step size.

$$p = e^{-\mu_t s} \rightarrow \frac{-\ln(p)}{\mu_t} = s$$

This gives a way to determine how far the particle travels before its next collision.

From there, the next step is to determine the scattering angle θ .

For light propagation, the Henyey Greenstein phase function[5] is used:

$$P(\theta) = \frac{1}{4\pi} \left(\frac{1 - g^2}{[1 + g^2 - 2g \cos(\theta)]^{3/2}} \right)$$

where g is an $\epsilon \in [-1, 1]$ and determines the probable direction of θ . The same method can be used by assigning P as an $\epsilon \in [0, 1]$. With that value and a chosen g , which will be derived below, we can solve for θ .

For ϕ , we assume a uniform distribution over $[0, 2\pi]$. This is justified by the symmetry of the spherical particle and the electromagnetic field interaction. At that point, the direction and step size of the particle are known, and the interaction can be tracked for any number of particles.

Finding g

For calculating g theoretically, we use the following:

$$g = \frac{2}{Q_{sca}} \sum_{n=1}^{\infty} \frac{n(n+2)}{n+1} \operatorname{Re}(a_n a_{n+1}^i + b_n b_{n+1}^i) + \frac{2n+1}{n(n+1)} \operatorname{Re}(a_n b_n^i)$$

where a_n and b_n are the TE and TM components derived from the interaction of a plane wave with a homogeneous isotropic sphere. To solve for them, we use the Hankel and Hankel Bessel functions ψ and ξ , respectively.

Let x be the cross sectional area of the particle over the wavelength of light:

$$x = \frac{\pi r^2}{\lambda}$$

$$a_n = \frac{m\psi_n(mx)\psi'_n(x) - \psi_n(x)\psi'_n(mx)}{m\psi_n(mx)\xi'_n(x) - \xi_n(x)\psi'_n(mx)}$$

$$b_n = \frac{\psi_n(mx)\psi'_n(x) - m\psi_n(x)\psi'_n(mx)}{\psi_n(mx)\xi'_n(x) - m\xi_n(x)\psi'_n(mx)}$$

From there, we can determine g .

Once g has been found, we have everything needed to propagate the particle in the desired way. The final step is to compare the model with the experiment. For this, we examined the change in intensity for each surface of the cuvette as the number of fat globules increased, and then compared those results with measurements from the same experimental conditions to evaluate the correlation.

Results & Data Analysis

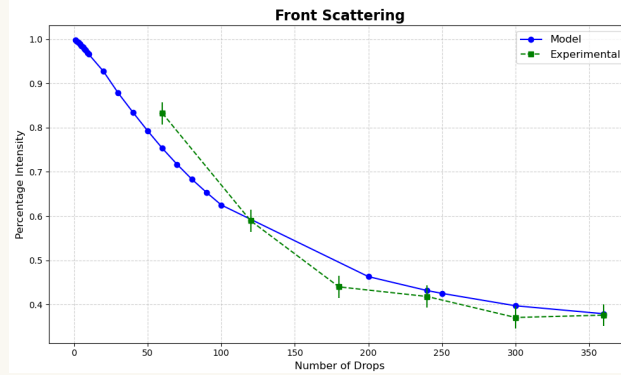


Figure 2: Foward Scattering Percentage as milk was added

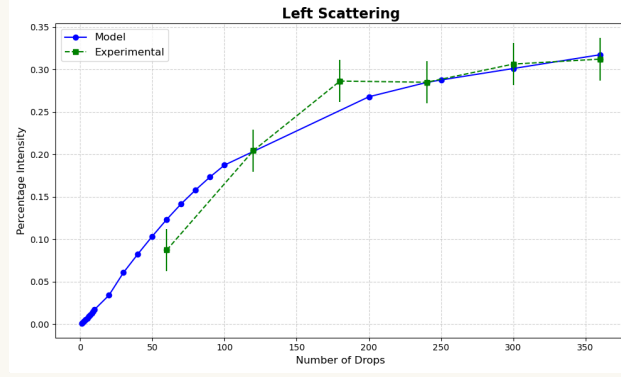


Figure 3: Left Scattering Percentage as milk was added

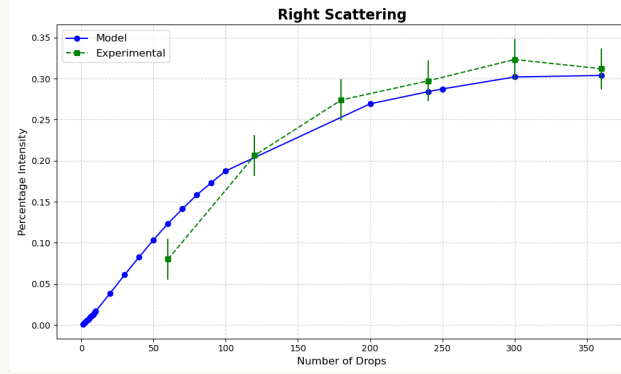


Figure 4: Right Scattering Percentage as milk was added

In total, 18 experimental points were collected for comparison with the simulation results. Based on the data, the largest general discrepancy between computation and experiment was roughly 8%, with most points falling within the greater than 2.5% error bar range. Taken together, these results suggest that the computational approach provides a reasonably good fit to the predicted light scattering behavior in the tested medium for the specific light source used.

Conclusion

This project investigated the accuracy of a Monte Carlo simulation coupled with Mie theory for modeling light scattering in whole milk, a turbid medium primarily influenced by fat globules. The simulation incorporated scattering and extinction coefficients derived from Mie theory and used the Henyey Greenstein phase function to model angular scattering behavior.

Experimental validation involved measuring the intensity of light from a 650 nm laser passing through a cuvette containing varying concentrations of milk, and then comparing those measurements with the simulation predictions. Despite experimental limitations, including Fresnel reflection losses and detector inefficiencies, the computational model showed relatively good agreement with the observed data, typically within a 7.9% margin, matching the expected uncertainty, and often within a 2.5% error range. These findings support the use of this simulation approach as a reasonable model for photon transport in similarly complex biological materials and provide a foundation for future applications in fields such as ophthalmology.

References

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